

**AIS-Watch: Detecting Anomalous Maritime Behavior and AIS Data
Manipulation using Deep Learning**



Final Year Project

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Abstract

The Automatic Identification System (AIS) is crucial for maritime safety and vessel tracking, but its vulnerabilities to manipulation, such as spoofing and intentional signal gaps, pose significant risks to maritime security and enable illicit activities. This project, "AIS-Watch," proposes a cost-effective, software-based solution leveraging machine learning to enhance maritime surveillance. By analyzing publicly available AIS data, the system employs an unsupervised Long Short-Term Memory (LSTM) autoencoder to model normal vessel trajectories. Deviations from these learned patterns, indicated by high reconstruction errors, are flagged as potential anomalies, specifically focusing on detecting AIS spoofing, anomalous signal interruptions, and other unusual behaviors. The primary outcome is an AI-driven proof-of-concept capable of automatically identifying suspicious maritime activities, thereby contributing to improved sea safety and the detection of illegal operations with minimal hardware dependency.

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Introduction

The Automatic Identification System (AIS) is a vital component of global maritime operations, mandated for most commercial vessels. It broadcasts dynamic information like position, speed, and course, along with static data such as vessel identity and type. While designed for collision avoidance and vessel traffic services, the reliance on AIS data has also exposed its vulnerabilities. Malicious actors can exploit these by intentionally turning off transponders ("going dark") or by transmitting falsified information ("spoofing") to conceal illicit activities such as illegal fishing, smuggling, sanctions evasion, or unauthorized rendezvous.

Traditional maritime surveillance often relies on costly hardware (radar, patrol vessels, aircraft) and intensive human monitoring, which may not be scalable or efficient enough to detect subtle AIS manipulations. This project aims to develop "AIS-Watch," an intelligent system that utilizes machine learning, specifically deep learning techniques, to analyze AIS data patterns. The core objective is to automatically detect abnormal vessel movements, with a particular emphasis on identifying sophisticated AIS data manipulation tactics like spoofing and suspicious signal gaps, using an unsupervised learning approach.

Existing System/Related Works

Current maritime surveillance incorporates a range of technologies and methods:

Radar and Satellite Imaging: Provide direct observation but can be expensive, limited by weather, and may require sophisticated analysis to correlate with AIS data.

Human Monitoring: Vessel Traffic Services (VTS) operators monitor AIS and radar feeds, but manual tracking of thousands of vessels is prone to oversight, especially for subtle anomalies.

Rule-Based Systems: Some existing systems use predefined rules (e.g., entry into restricted zones, excessive speed) to flag alerts. While useful, they may struggle with novel or complex anomalous behaviors and can generate many false positives.

Early Machine Learning Approaches: Several studies have applied machine learning to AIS data for anomaly detection. Techniques include clustering (e.g., DBSCAN to find vessels deviating from common routes), Support Vector Machines (SVMs), and Random Forests. While effective for certain types of anomalies, some require labeled datasets (which are scarce for illicit activities) or may not be optimized for capturing the complex temporal dependencies inherent in vessel trajectories or detecting deliberate AIS manipulation. Recent research (e.g., by Zhang et al., Xie et al., as per a 2025 review on Transformer-based AIS monitoring) explores advanced models like Transformers and LSTMs for trajectory prediction and behavior detection, indicating a trend towards more sophisticated sequential data analysis. However, a dedicated, cost-effective, and unsupervised approach focusing specifically on robust spoofing and gap detection using LSTM autoencoders offers a valuable contribution.

The proposed system builds on these advancements by focusing on an unsupervised deep learning method that learns normal patterns in-depth and is thus sensitive to various deviations, including deliberate data falsification, without requiring pre-labeled anomaly data.

Proposed System

"AIS-Watch" is an AI-based system designed to detect anomalous maritime behavior by analyzing patterns in historical and near-real-time AIS data. The core of the system is an unsupervised Long Short-Term Memory (LSTM) autoencoder model.

The system operates as follows:

It ingests and preprocesses large volumes of AIS data, creating standardized vessel trajectories.

The LSTM autoencoder is trained exclusively on trajectories deemed "normal." The encoder part of the model learns to compress a trajectory into a compact latent representation, and the decoder learns to reconstruct the original trajectory from this representation.

During operation, new vessel trajectories are fed into the trained autoencoder.

The system calculates the reconstruction error (e.g., Mean Squared Error) between the input trajectory and the model's reconstructed output.

Trajectories with reconstruction errors significantly higher than those observed for normal data during training are flagged as anomalous. This high error indicates that the input trajectory deviates from the learned patterns of normality, potentially due

to spoofing, an unusual signal gap, erratic movement, or other suspicious behavior.

This unsupervised approach is particularly suited for this problem due to the scarcity of labeled data for illicit maritime activities.

Scope

The system will exclusively use publicly available AIS datasets.

No specialized hardware sensors like radar or LIDAR will be integrated for data collection in this project phase.

Anomaly detection will focus on:

AIS location spoofing (e.g., "teleportation," route masking).

Anomalous AIS signal gaps ("going dark" in suspicious contexts).

Highly unusual kinematic behaviors (e.g., loitering in unexpected areas, erratic speed/course changes inconsistent with vessel type).

The output will be a proof-of-concept model and a system capable of identifying and flagging potential anomalies for further investigation.

Real-time, instantaneous detection is a future goal; this project will focus on batch processing of historical or near-current data.

Proposed System Modules

Module 1: AIS Data Ingestion and Trajectory Intelligence Engine

Techniques: Data Sourcing (from public archives like MarineCadastre.gov, Kaggle), Data Cleaning (handling missing values, removing duplicates, correcting obvious errors), Advanced Feature Engineering (calculating kinematic features like speed, course, rate of turn, acceleration; deriving contextual features if simple GIS data is integrated e.g., distance to port; engineering specific indicators for potential spoofing like implausible location jumps, and features quantifying AIS signal gaps), Trajectory Segmentation (grouping AIS messages into coherent vessel tracks based on MMSI and time continuity). This module prepares high-quality, feature-rich sequential data for the learning model.

Module 2: Unsupervised Anomaly Detection Model (LSTM Autoencoder)

Techniques: Deep Learning Model Architecture (designing an LSTM-based autoencoder with an encoder to learn a compressed representation of trajectories and a decoder to reconstruct them), Unsupervised Training (training the autoencoder on a large dataset of "normal" vessel trajectories to learn a robust representation of normality, using a reconstruction loss like Mean Squared Error), Hyperparameter Tuning (optimizing model parameters like number of layers, hidden units, learning rate using a validation set of normal data). This module creates the core intelligence that understands "normal" maritime movement.

Module 3: Anomaly Identification, Scoring, and Evaluation

Techniques: Inference and Reconstruction Error Analysis (feeding new/unseen trajectories to the trained model and calculating their reconstruction error), Thresholding (establishing a dynamic or static threshold on the reconstruction error to distinguish between normal and anomalous trajectories), Anomaly Flagging & Visualization (marking trajectories exceeding the threshold as anomalous and enabling their visualization on a map for review), Model Evaluation (assessing performance using metrics like reconstruction error distributions for normal vs. synthetic anomalous data; if using synthetic anomalies: precision, recall, F1-score. Qualitative review of flagged anomalies). This module applies the learned model and provides actionable outputs.

Data Collection and Preprocessing

Data will be sourced from publicly accessible AIS repositories such as:

MarineCadastre.gov (U.S. waters)

NOAA Office for Coastal Management (AccessAIS tool)

Kaggle AIS datasets

Academic datasets (e.g., from GitHub repositories like eyesofworld/Maritime-Monitoring)

The preprocessing pipeline will involve:

Parsing: Reading raw AIS data (typically CSV or NMEA, though CSV is preferred for simplicity).

Cleaning:

Removing records with invalid essential fields (MMSI, timestamp, latitude, longitude).

Handling missing values: Interpolation for very short gaps in continuous tracks, otherwise, long gaps might be features themselves.

Filtering out duplicate messages.

Correcting obvious errors (e.g., coordinates on land, impossible speeds based on vessel type if this information is available and reliable).

Feature Engineering:

Kinematic: SOG, COG, calculated Rate of Turn (ROT), acceleration.

Trajectory-level: Duration of track segment, total distance.

AIS Gap Features: Duration of signal loss between consecutive pings from the same MMSI.

Spoofing Indicators (Heuristic): Max speed between pings (distance/time), sudden large jumps in lat/lon.

Trajectory Segmentation: Grouping AIS messages by MMSI and sorting by timestamp. A trajectory can be defined as a sequence of AIS messages from a single vessel within a continuous period of activity or a specific voyage leg.

Normalization: Scaling numerical features to a common range (e.g., [0, 1]) for optimal neural network performance.

Model Development

The core of the anomaly detection system will be an LSTM-based Autoencoder.

Architecture:

Encoder: Will consist of one or more LSTM layers followed by a dense layer to compress the input sequence (trajectory) into a fixed-size latent vector.

Latent Space: The compressed representation capturing the essential characteristics of the trajectory.

Decoder: Will consist of one or more LSTM layers (often mirroring the encoder structure) that take the latent vector as input and attempt to reconstruct the original trajectory sequence. RepeatVector might be used to feed the latent vector to each time step of the decoder LSTM.

Activation Functions: tanh or relu for LSTMs, sigmoid or linear for the output layer

depending on normalization.

Training:

The model will be trained in an unsupervised manner using only normal vessel trajectories.

The loss function will be a measure of reconstruction error, typically Mean Squared Error (MSE), between the input trajectory and the output of the decoder.

An optimizer like Adam will be used to minimize this loss.

Rationale: LSTMs are chosen for their ability to capture temporal dependencies in sequential AIS data. The autoencoder framework allows for unsupervised learning, which is crucial given the lack of large, labeled datasets of anomalous maritime behavior.

Potential Enhancement: Attention mechanisms could be explored within the LSTM structure to help the model focus on more salient parts of the trajectory during reconstruction, potentially improving sensitivity to subtle anomalies.

Evaluation and Validation

Reconstruction Error Analysis: The primary evaluation will involve analyzing the distribution of reconstruction errors on a held-out validation set of normal data and on any available or synthetically generated anomalous data. A clear separation in distributions would indicate good model performance.

Threshold Selection: The validation set of normal data will be used to determine an appropriate threshold for the reconstruction error. Anomalies will be flagged if their error exceeds this threshold (e.g., mean error on normal data + $N \times$ standard deviation, or a specific percentile).

Metrics (with Labeled/Synthetic Data): If synthetic anomalies are generated or a small labeled test set is available:

Precision: Proportion of flagged anomalies that are true anomalies.

Recall: Proportion of true anomalies that are correctly flagged.

F1-Score: Harmonic mean of precision and recall.

ROC Curve and AUC: To evaluate the trade-off between true positive rate and false positive rate.

Qualitative Validation: Subject matter experts (if available) or visual inspection of flagged anomalous trajectories plotted on a map will be used to assess the real-world relevance and plausibility of the detected anomalies. This helps identify

if the model is picking up genuinely suspicious behavior or just noise.

Case Studies: Analyzing specific known (if any public data exists) or highly suspicious patterns to see if the model correctly identifies them.

Advantages of the Proposed System

Unsupervised Learning: Does not require large, labeled datasets of illicit activities, which are difficult and costly to obtain.

Detection of Novel Anomalies: Can potentially detect previously unseen types of anomalous behavior as they will likely deviate from learned normal patterns.

Focus on Temporal Dynamics: LSTMs are well-suited to model complex time-series data, capturing subtle sequential patterns indicative of spoofing or unusual maneuvers.

Cost-Effective: Primarily a software-based solution using publicly available data, minimizing hardware and operational costs compared to traditional surveillance.

Automation: Reduces the need for constant manual monitoring, allowing human analysts to focus on investigating high-priority alerts.

Scalability: Deep learning models can scale to handle large volumes of AIS data.

Limitations/Disadvantages

Dependence on AIS Data Quality: The system's performance is contingent on the availability and quality of AIS data. Errors or biases in the input data can affect results.

Defining "Normal": The model's understanding of normality is based on its training data. If the training data inadvertently contains certain types of anomalies, the model might learn them as normal. Careful curation of training data is essential.

Threshold Setting: Determining the optimal reconstruction error threshold can be challenging and may require iterative tuning.

Interpretability: Deep learning models, including LSTMs, can be "black boxes," making it sometimes difficult to understand precisely why a specific trajectory was flagged. Techniques like LIME or SHAP, or analyzing attention weights (if used), can be explored for better interpretability but add complexity.

False Positives/Negatives: Like any detection system, it may produce false alarms or miss some actual anomalies.

Computational Resources for Training: Training deep learning models can be computationally intensive, though cloud-based GPU resources can mitigate this.

Applications

Maritime Security: Detecting vessels involved in piracy, smuggling, terrorism, or sanctions evasion.

Fisheries Monitoring: Identifying illegal, unreported, and unregulated (IUU) fishing activities by detecting vessels operating in closed areas, using prohibited gear patterns, or "going dark."

Environmental Protection: Monitoring for vessels potentially involved in illegal dumping or pollution.

Port Security: Identifying anomalous vessel movements near critical infrastructure.

Search and Rescue: Providing additional data points or identifying unusual behavior prior to a distress event.

Traffic Management: Enhancing Vessel Traffic Service (VTS) capabilities by automatically highlighting vessels exhibiting risky behavior.

SDG Goals

This project aligns with several UN Sustainable Development Goals (SDGs):

SDG 14: Life Below Water: By helping to combat IUU fishing, the project contributes to the conservation and sustainable use of marine resources.

SDG 16: Peace, Justice and Strong Institutions: By providing tools to detect and deter illicit maritime activities, the project supports efforts to reduce illicit financial and arms flows, and combat transnational organized crime.

SDG 9: Industry, Innovation, and Infrastructure: The project represents an innovative application of AI technology to improve maritime infrastructure monitoring and safety.

Hardware/Software Requirements

Emphasis is placed on cost-effectiveness and leveraging readily available resources.

Hardware (Minimal & Cost-Effective):

Software:

Operating System: Windows 10/11, Ubuntu Linux (20.04+), or macOS (latest versions).

Programming Language: Python (3.8 or newer).

Key Python Libraries:

Pandas: For data manipulation and analysis.

NumPy: For numerical operations.

Scikit-learn: For preprocessing (e.g., scaling) and evaluation metrics.

TensorFlow (with Keras API) or PyTorch: For building and training the LSTM autoencoder.

Matplotlib, Seaborn: For plotting and data visualization.

Folium: For visualizing trajectories on interactive maps.

(Optional) GeoPandas: If performing advanced geospatial feature engineering.

IDE: Jupyter Notebook/JupyterLab, VS Code with Python extension, or PyCharm Community Edition.

Version Control: Git and a GitHub repository for code management and collaboration.

References

(This section would typically include specific citations. For this template, general categories of references are provided. In an actual report, you would list specific papers, dataset sources, and software documentation.)

Documentation for AIS data sources (e.g., MarineCadastre.gov, NOAA AccessAIS).

Relevant academic papers on AIS data analysis, anomaly detection in maritime surveillance, and applications of LSTMs or autoencoders in time-series anomaly detection. (e.g., papers found in searches related to "AIS anomaly detection machine learning," "LSTM autoencoder for trajectory anomaly detection").

Papers on AIS vulnerabilities and manipulation techniques.

Documentation for Python libraries used (Pandas, NumPy, Scikit-learn, TensorFlow/PyTorch, etc.).

United Nations Sustainable Development Goals official website/documents.

[Specific relevant studies like those by Minßen et al. or the review by Xie et al. (2025) on Transformer-based AIS monitoring, if details were accessible and used for methodological inspiration.]